

Shaik Abdus Sattar

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Education

Amrita Vishwa Vidyapeetham

Bachelor of Engineering in Computer Science; CGPA: 8.4/10.0

Coimbatore, Tamil Nadu

Aug 2023 - Present

Excellencia Junior College

Intermediate/+2 (State Board): 98%; JEE-Mains: 95.87 percentile

Hyderabad, Telangana

2023

Technical Skills

Programming Languages: Python, C++, Java, SQL, JavaScript, Haskell

Machine Learning & AI: Deep Learning, CNNs, Transformers, NLP, Computer Vision

ML Frameworks & Data: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn

Full-Stack Development: Node.js, React, Next.js, Tailwind CSS, Express.js, MongoDB

Cloud & Tools: Git, AWS, REST APIs, Data Analysis, Data Visualization

Research Experience

Clinically-Informed Deep Learning for Radiotherapy QA Prediction

2025 - Present

Research Assistant | Advisors: [Dr. Amit Agarwal](#), [Dr. Rose Kamal](#)

- Developing CNN and hybrid deep learning models to predict pre-treatment Gamma passing rates from dose maps and Treatment Plan Complexity Metrics; designing clinically weighted loss functions and structured pipelines for automated radiotherapy QA.

Efficient 3D Molecular Graph Kernels for Molecular Property Prediction

2026 - Present

Research Assistant | Advisor: [Dr. T. Ramraj](#)

- Developing efficient 3D graph kernel frameworks incorporating bond lengths, angles, and torsional features for molecular similarity modeling; investigating scalable 2D-to-3D kernel strategies to reduce time complexity.
- Designing efficient data structures to reduce the computational complexity of 3D graph kernel computations from approximately $O(n^6)$ to $O(n^2)$, improving scalability for large-scale molecular property prediction.

Projects

Orbit Wars: Multi-Agent Deep RL AI | PyTorch, PPO, PBT, Transformers



- Engineered a custom deep reinforcement learning architecture combining spatial Transformers and Global LSTMs to process dynamic temporal states in a real-time strategy physics game.
- Designed a highly scalable training pipeline utilizing Proximal Policy Optimization (PPO) and Population-Based Training (PBT) to orchestrate concurrent self-play workers and dynamic Elo matchmaking.

LFW Face Generation & Latent Space Editing | AutoEncoders, VAE, Latent Interpolation, PyTorch



- Trained VAE models on LFW dataset for photorealistic face synthesis with latent space interpolation.
- Implemented attribute-driven latent vector editing to analyze disentanglement and geometry of the learned generative manifold.

HASYv2 Handwritten Math Symbol Classification | PyTorch, CNN, BatchNorm, Data Augmentation



- Built PyTorch CNN for 369-class symbol classification with custom preprocessing, BatchNorm, and adaptive pooling.
- Won 1st place in a university competition achieving 84% accuracy via hyperparameter tuning under constraints.

Certifications

AWS Cloud Practitioner Essentials - AWS (2025)

Deep Learning Specialization - DeepLearning.ai (2025)

Machine Learning Specialization - Stanford Online (2025)

Achievements & Leadership

Head Coordinator - Gokulashtami Event (2025)

Hackathon Winner - IETE Hackathon (2025)

AIR 270 - AEEE Exam (2023)

1st Place - Western Group Music, Gokulashtami (2025)